

Avantika Bansal

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 LinkedIn |  GitHub |  Portfolio



EDUCATION

BTech - Biological Systems Engineering	Plaksha University, Mohali, India	8.93	2027
12th - CBSE	MGD Girls' School, Jaipur, India	94.8%	2023
10th - CBSE	MGD Girls' School, Jaipur, India	93.2%	2021

RESEARCH EXPERIENCE

Research Intern, Amgen Scholars Program Jun 2026 - Aug 2026
Computational Aptamer Optimization for Electrochemical Biosensor | *Dr. Abhishek Srivastava, IIIT-H*

- Established a **ZnO nanorod based electrochemical sensor** for rapid detection of **Tuberculosis**.
- Fabricated and characterized **ZnO nanorod-modified IDEs using hydrothermal synthesis** for electrochemical biosensing.
- Developed a computational pipeline for aptamer optimization using **2D and 3D modelling** with RNA structure simulation tools.
- Extracted **sequence, structural, physicochemical, docking features and binding affinity** for aptamer-protein pairs.
- Presented at the **Amgen Asia Symposium**, University of Tokyo, engaging with 75+ researchers from across Asia and Australia.

Assessing Fungal Microbiome to Investigate Soil Carbon Sequestration | *Dr. Navjot Kaur* Jul 2025 - Present

- Characterized **3 economically affecting fungal pathogens** on a molecular level for species level identification.
- Performed fungal culturing, microscopy, genomic DNA extraction and PCR amplification for species identification.
- Studied **host-pathogen interaction** of the fungal species to understand pathogen life cycle.
- Analyzed fungal sequences using **phylogenetic trees with MEGA** alignment tool to validate species identity and similarity.
- Conducted a comparative analysis for **ITS and species specific primers** obtaining 95% alignment with reference sequences.

Algal Facades for Sustainable Buildings | *Prof. Malini Balakrishnan, Prof. Prashanth Kumar* Jan 2025 - May 2026

- Developed and analyzed efficiency of **algal photobioreactors** to evaluate energy-saving potential.
- Ran **15+ building-energy simulations** varying facade parameters with DesignBuilder software to model performance.
- Cultured algae and carried out media preparation, inoculation, sampling, and growth-curve analysis to characterize algal growth.
- Designed and operated a lab scale photobioreactor to **experimentally validate algal growth and nutrient absorption**.
- Conducted **life-cycle assessment (LCA)** using OpenLCA to evaluate environmental impacts of the algal facade vs blinds.
- Found energy savings value of upto **-20 Global Warming Potential** units for facade operation.

PUBLICATIONS

Bansal, P., Bansal, A., Kaur, N. (2026). *Molecular characterization of Fusarium oxysporum, Alternaria alternata, and Sclerotium rolfsii: A comparative assessment of ITS and species-specific markers*. Submitted for review in the Stanford Undergraduate Research Journal (SURJ)

Bansal, A., Mathai, B., Jadhav, N., Balakrishnan, M., Kumar, P. S. (2025). *Designing an Algal Façade for Improving Energy Efficiency in a Building in North India*. Oral presentation at the X International Conference on Sustainable Energy and Environmental Challenges (X SEEC)

PROJECTS

Deep Learning-Enhanced Metabolic Control for PHA Production | *Prof. Monika Sharma* Aug 2026 - Present

- Building a computational framework to optimize the **production of bioplastic (PHA)** in Pseudomonas putida.
- Developing **metabolic sub-network reconstruction** for enhancing PHA production pathways.
- Formulating a **dynamic simulations and MCA** based framework to analyze metabolic engineering targets.

Genetic Engineering for Cadmium Phytoremediation | *Prof. Arshdeep Sidhu* Mar 2026 - May 2026

- Designed a genetic engineering strategy for *Populus x canescens* to **enhance cadmium uptake and stress tolerance**.
- Developed a multi-gene construct targeting **PcNRAMP1, GSH1/GSH2, SOD, and APX** for enhanced phytoremediation.
- Proposed framework increases **cadmium uptake more than 50%** based on hypothetical estimates.

Multi-Agent RL for Linear Salp-Chain Underwater Navigation | *Prof. Sandeep Manjanna* Jan 2026 - May 2026

- Developed a **control policy** for a linear chain of underwater robots for coordinated obstacle avoidance navigation.
- Worked with a **2D underwater navigation** RL environment with salp chain agents from GRASP Lab, UPenn.
- Implemented **multi-agent policies** IQL, MAPPO, and MADDPG for target navigation and obstacle avoidance.
- Designed **dynamic underwater obstacle-placement curriculum**, achieving an **81%** collision-free navigation success rate.

Inverse PINN Modelling of Flow in a Collapsible Compliant Tube | *Prof. Shashikant Pawar* Aug 2026 - Present

- Analyzing flow–structure interactions in collapsible tubes to study **pressure and velocity fields** in venous blood flow.
- Constructing **fluid–structure interaction (FSI) models** and experimental setup for tube deformation and flow measurements.
- Implementing a **physics-informed inverse model (PINN)** for validating experimental measurements.

ZnO-NiO Nanocomposite Synthesis | *Prof. Chaitanya Lekshmi Indira* Mar 2025 - May 2025

- Fabricated and analyzed a 1:1 **ZnO-NiO nanocomposite** using an Ultrasonication-Assisted Mechanical method.
- Characterized samples using **XRD, FTIR, and UV-Vis spectroscopy** and obtained distinctive intensity and absorption peaks.
- Achieved up to **90% similarity** between sample results and characteristic properties of lab-grade chemicals.

SKILLS

- **Biosystems:** Microbial culture, molecular cloning, DNA extraction, qPCR amplification, electrochemical analysis
- **Programming Languages:** C, C++, Python
- **Computational Tools:** BLAST, SnapGene, ClustalW, MEGA; DesignBuilder, COMSOL, OpenLCA

RELEVANT COURSEWORK

BE2007: Bioprocess Engineering	Grade: A	BE3012: Cell Biology	Grade: A-
BE3007: Genetics and Genetic Engineering	Grade: A-	BE2002: Biochemistry and Molecular Biology	Grade: A-
BE3001: Stochastic Modelling in Biology	Grade: A-	BE2005: Material Science for Bioengineering	Grade: A-
BE3014: Lab-on-a-Chip Devices	Grade: A-	ME3002: Fluid Mechanics	Grade: A-
RO3004: Reinforcement Learning	Grade: A-	AI3011: Machine Learning	Grade: A

LEADERSHIP ROLES

Cohort Representative | Biological Systems Engineering, Plaksha University Jan 2025 – Present

- Representing BSE cohort, facilitating communication between 20+ faculty members and the student body.
- Bridging student-faculty communication on course policies resolving student academic challenges through continuous feedback.

President | Nebula, Plaksha Well-Being Club Feb 2025 – Present

- Leading a 30 member team to design and implement student-centric mental health and stress-relief programs.
- Conducted events and competitions to contribute to a more supportive and resilient campus culture.

Core Member, Organizing Committee | Dotslash Hackathon, Plaksha University Nov 2024

- Collaborated on logistics, venue layout, and challenge-track design for a 120+ participant inter-college hackathon.

House Captain | MGD Girls’ School Jul 2022 – Apr 2023

- Led 500+ students in cultural and sports events, securing the award for best house in academics and co-curricular activities.
- Coordinated major annual events for 1000+ students as part of a 37-member student council.

ACHIEVEMENTS

- **Runners-up, Hack Plaksha** (Fintech Track, Feb 2024) - Developed Dhanshakti, a financial management app to empower rural self-help groups by simplifying savings and loan tracking.
- **Gold Medal, District Karate Tournament** - Secured first place in Individual Shihai in Under 18 category.
- **Merit Scholarship, Plaksha University:** Recognized for academic excellence with a GPA above 8 and extracurricular.
- Completed Foundation certification for **BS in Data Science** from IIT Madras.